

Functional Annotation Report: *tauD* (Q88RA3) in *Pseudomonas putida* KT2440

Gene: *tauD* (synonym *atsK*; ordered locus PP_0230) **Protein:** Alpha-ketoglutarate-dependent taurine dioxygenase (TauD) **UniProt:** Q88RA3 · **EC:** 1.14.11.17 · **Length:** 277 aa · **Organism:** *Pseudomonas putida* KT2440 (PSEPK)

1. Summary (Answer to the Research Question)

tauD encodes a soluble, cytoplasmic **Fe(II)/ α -ketoglutarate (2-oxoglutarate)-dependent taurine dioxygenase** (EC 1.14.11.17). Its primary function is to catalyze the O_2 - and α -ketoglutarate-dependent **hydroxylation of taurine (2-aminoethanesulfonate)** at its C1 carbon; the resulting unstable 1-hydroxy intermediate spontaneously decomposes to release **inorganic sulfite** plus aminoacetaldehyde, while the co-substrate α -ketoglutarate is oxidatively decarboxylated to succinate and CO_2 . The enzyme's biological role is **organosulfonate $\rightarrow$ sulfur assimilation**: it liberates sulfite for cysteine biosynthesis when preferred sulfur sources (sulfate, cysteine) are scarce. This exact protein (TauD_Pp, PP_0230) has been directly characterized biochemically, kinetically, and crystallographically (Knauer et al., 2012,).

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2. Identity Verification (Critical Check)

The target was cross-checked against UniProt Q88RA3 and the primary literature:

- **Gene symbol match:** UniProt lists `Name=tauD`, `Synonyms=atsK`, `OrderedLocusNames=PP_0230`. The recommended protein name is "Alpha-ketoglutarate-dependent taurine dioxygenase," consistent with the target description.
- **Organism match:** *Pseudomonas putida* KT2440 confirmed.

- **Family/domain match:** Belongs to the **TfdA/TauD dioxygenase family** (Pfam PF02668 "TauD"; InterPro IPR003819 TauD/TfdA-like, IPR042098 TauD-like superfamily). This matches the enzymology in the literature.
- **Direct experimental confirmation:** Knauer et al. (2012) explicitly purified, kinetically characterized, and solved three apo crystal structures of the taurine dioxygenase **from *P. putida* KT2440 (TauD_Pp)** — i.e., this exact protein — and showed it closely mirrors *E. coli* TauD in spectra, structure, and kinetics

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Note on the *atsK* synonym (potential source of confusion): The genome-annotation synonym *atsK* refers to a paralogous 2-oxoglutarate-dependent **alkylsulfatase** biochemically characterized in a *different* strain, *P. putida* **S-313** (Kahnert & Kertesz, 2000,

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Müller et al., 2004/2005,

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15542595). *AtsK* cleaves alkyl **sulfate esters** (C4–C12), whereas the KT2440 PP_0230 product was directly shown to be a bona fide **taurine sulfonate** dioxygenase. The two are homologous members of the same superfamily (~38% identity between *AtsK* and *E. coli* TauD) with the same fold and chemistry, but distinct substrate specificity. The *AtsK* work is therefore used here for **mechanistic/structural inference**, while function of Q88RA3 itself rests on the KT2440-specific study.

3. Primary Function: Reaction and Substrate Specificity

Reaction catalyzed (EC 1.14.11.17)



TauD is a **dioxygenase** that splits molecular O₂: one oxygen atom is incorporated into succinate (via oxidative decarboxylation of α-ketoglutarate) and the other is used to hydroxylate the substrate. Hydroxylation occurs at **C1 of taurine**, producing an unstable 1-hydroxy-2-aminoethanesulfonate that decomposes non-enzymatically to **sulfite** +

aminoacetaldehyde (UniProt FUNCTION, ECO:0000269

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Knauer et al. 2012).

Substrate specificity

- **Primary substrate:** taurine (2-aminoethanesulfonate), a naturally abundant aminosulfonate. TauD "preferentially liberates sulfite from taurine" (van der Ploeg et al., 2001,

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- **Co-substrate:** α -ketoglutarate (2-oxoglutarate), consumed stoichiometrically and converted to succinate + CO₂.

- **Cofactor:** one catalytic **Fe(II)** ion per subunit (UniProt COFACTOR,

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activity requires O₂ and is stimulated by ascorbate (a common feature of this family that re-reduces adventitiously oxidized iron).

- The homologous alkylsulfatase AtsK, by contrast, accepts **alkyl sulfate esters** and even a range of α -keto-acid co-substrates, illustrating how small active-site differences retune substrate scope within this superfamily (Kahnert & Kertesz, 2000,

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4. Catalytic Mechanism (Structural & Spectroscopic Evidence)

TauD is the **paradigm system** for the Fe(II)/ α KG dioxygenase superfamily:

- **Active-site iron** is held by the canonical **2-His-1-carboxylate "facial triad"** (Nam et al., 2014,

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Residue-level mapping of Q88RA3 (from its crystal structures, UniProt feature table) assigns this triad to **His97, Asp99 and His253** (the diagnostic HXD...H motif). The co-substrate 2-oxoglutarate is coordinated by **Thr124, His253 (bidentate to Fe), Arg264 and Arg268**, and the **taurine/sulfonate pocket** is formed by **His68, Tyr71, Asn93, Val100 and Arg268** —

providing a structural basis for taurine substrate specificity. Three apo crystal structures of this exact protein are deposited: **PDB 3PVJ, 3V15, 3V17** (Knauer et al., 2012,

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- **Catalytic cycle:** Fe(II) binds α -ketoglutarate and then O_2 ; oxidative decarboxylation of α -ketoglutarate produces CO_2 , succinate, and a highly reactive **high-valent Fe(IV)=O (ferryl) intermediate**. TauD was the **first enzyme in which a mononuclear nonheme iron(IV)-oxo intermediate was spectroscopically characterized (2003)** (Nam et al., 2014,

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The ferryl species abstracts a hydrogen atom from taurine C1, and oxygen rebound yields the hydroxylated product.

- **Self-inactivation control:** In the closely related AtsK, non-productive O_2 activation is quenched by **self-hydroxylation of an active-site tyrosine (Tyr168)** that coordinates iron — a built-in protective mechanism against harmful side reactions (Müller et al., 2005,

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This structural insight extends to TauD-family enzymes generally.

- **Fold:** The AtsK/TauD fold is a double-stranded β -helix (cupin/jelly-roll) "distorted barrel" typical of the superfamily; the AtsK structure is "closely related to that of the taurine/ α -ketoglutarate dioxygenase TauD" (Müller et al., 2004,

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5. Quaternary Structure

The *P. putida* KT2440 TauD is a **homotetramer (dimer of dimers)**, binding one Fe(II) per subunit (UniProt SUBUNIT/COFACTOR,

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Knauer et al. 2012 title: the taurine dioxygenases from *P. putida* and *E. coli* "are tetramers"). This oligomeric state — shared with the homotetrameric AtsK (Kahnert & Kertesz, 2000,

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— is unusual for the superfamily, whose members are more often monomeric or dimeric.

6. Subcellular Localization

TauD is a **cytoplasmic, soluble enzyme**. Evidence/inference: - **Bioinformatic check (this work)**: the Q88RA3 sequence carries **no signal peptide and no transmembrane segment** (no Sec/Tat export or membrane-anchoring features), and the N-terminus (MSLTITPLSPALGA...) is not a cleavable signal — consistent with a soluble cytoplasmic protein. - The protein was purified as a soluble crystallizable tetramer (P 22221834),

consistent with cytoplasmic residence. - **Pathway logic**: Taurine is first imported across the inner membrane by the **TauABC ABC transporter**, after which the intracellular TauD desulfonates it (van der Ploeg et al., 2001, P 11479697).

Desulfonation therefore occurs **inside the cell, in the cytoplasm**, where the liberated sulfite feeds the cysteine-biosynthetic (sulfate assimilation) branch.

7. Pathway and Physiological Context

- **Pathway**: Aerobic **organosulfonate (taurine) catabolism for sulfur assimilation**. TauD provides the desulfonation step that converts an organic sulfonate into inorganic **sulfite** (SO_3^{2-}), which is reduced to sulfide and incorporated into cysteine.
- **Genetic organization / regulation**: In the model *E. coli* system, *tauD* is the fourth gene of the **tauABCD operon** (TauABC = ABC importer; TauD = desulfonase). Expression is induced under **sulfate/cysteine starvation** and requires the **LysR-type regulators CysB (master sulfur regulator) and Cbl** (van der Ploeg et al., 2001, P 11479697).

The homologous *P. putida* α KG-dependent sulfonatase system (AtsK) is likewise "only expressed under sulfur starvation conditions, providing a selective advantage for bacterial growth in soils and rhizosphere" (Müller et al., 2004, P 15023059).

- **Strain-specific genomic evidence (this work):** In the *P. putida* KT2440 genome, *tauD* (PP_0230) is directly clustered with a complete taurine ABC-transporter operon — **PP_0231 *tauC*** (permease), **PP_0232 *tauB*** (ATP-binding), **PP_0233 *tauA*** (periplasmic taurine-binding protein) — i.e. the PP_0230–PP_0233 module recapitulates the *E. coli tauABCD* architecture. Immediately adjacent lies the parallel **ssuEADCB** cluster (PP_0236–PP_0240: SsuE FMN reductase, SsuA binding protein, SsuD alkanesulfonate monooxygenase, SsuC permease, SsuB ATPase) for FMNH₂-dependent desulfonation of other aliphatic sulfonates, and the master regulator **CysB** is encoded elsewhere (PP_2327). This confirms — *in the actual target organism* — that TauD works within an intact import-and-desulfonation module rather than by inference from *E. coli* alone: periplasmic **TauA** captures taurine, **TauBC** import it across the inner membrane, and cytoplasmic **TauD** liberates sulfite.
 - **Ecological role:** For a soil/rhizosphere organism like *P. putida* KT2440, the ability to mine sulfur from abundant environmental sulfonates (taurine and related compounds) when inorganic sulfate is limiting is a competitive advantage.
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8. Supported and Refuted Hypotheses

Hypothesis	Verdict	Basis
Q88RA3 is a taurine/αKG dioxygenase (EC 1.14.11.17)	Supported	UniProt + direct characterization of KT2440 TauD (P 22221834)
Reaction yields sulfite + aminoacetaldehyde (+ succinate + CO ₂)	Supported	UniProt catalytic activity; (P 22221834)
Uses mononuclear nonheme Fe(II), 2-His-1-carboxylate triad, ferryl intermediate	Supported	(P 24524675); superfamily mechanism
Homotetramer, 1 Fe(II)/subunit	Supported	UniProt; (P 22221834)
Functions cytoplasmically in sulfur-starvation-induced sulfonate assimilation	Supported	Pathway logic + (P 11479697) (P 15023059)
KT2440 possesses an intact tauABCD-type import/desulfonation module around PP_0230	Supported	Genomic clustering PP_0230–PP_0233 (tauD-tauCBA) + adjacent ssuEADCB, this work
The gene product is the S-313 alkylsulfatase AtsK (alkyl sulfate esters)	Refuted for this protein	AtsK is a paralog from a different strain; KT2440 PP_0230 was shown to be a taurine dioxygenase (P 22221834). <i>atsK</i> is only a genome-annotation synonym.

9. Limitations and Future Directions

- **Substrate range not exhaustively mapped for KT2440 TauD:** while taurine is the confirmed physiological substrate, quantitative kinetics for alternative sulfonates/ α -keto-acids in the KT2440 enzyme specifically were not fully detailed; broader specificity is inferred from *E. coli* TauD and *P. putida* AtsK.
- **Localization is inferred** (soluble/cytoplasmic) from sequence features and pathway logic rather than an explicit fractionation experiment for Q88RA3.
- **Direct regulatory data in KT2440** (CysB/Cbl control of PP_0230) is extrapolated from *E. coli*; although the *tauABCD* and *ssuEADCB* gene clusters and *cysB* are all present in the KT2440 genome (this work), strain-specific transcriptional/promoter confirmation would further strengthen the model.
- Future work: substrate-bound/holo crystal structures of TauD_Pp, in-vivo growth phenotypes of a KT2440 *tauD* deletion on taurine as sole sulfur source, and quantitative regulon mapping.

Key References

- Knauer SH, Hartl-Spiegelhauer O, Schwarzingner S, Hänzelmann P, Dobbek H. *The Fe(II)/ α -ketoglutarate-dependent taurine dioxygenases from Pseudomonas putida and Escherichia coli are tetramers.* FEBS J, 2012.

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(Direct characterization of the target protein.)

- van der Ploeg JR, Eichhorn E, Leisinger T. *Sulfonate-sulfur metabolism and its regulation in Escherichia coli.* Arch Microbiol, 2001.

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(Pathway and regulation.)

- Nam W, Lee YM, Fukuzumi S. *Tuning reactivity and mechanism in oxidation reactions by mononuclear nonheme iron(IV)-oxo complexes.* Acc Chem Res, 2014.

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(Ferryl intermediate; 2-His-1-carboxylate triad.)

- Kahnert A, Kertesz MA. *Characterization of a sulfur-regulated oxygenative alkylsulfatase from Pseudomonas putida S-313*. J Biol Chem, 2000.

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(Homologous AtsK; substrate scope; tetramer.)

- Müller I, et al. *Crystal structure of the alkylsulfatase AtsK...* J Mol Biol, 2004.

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(Fold, mechanism, sulfur-starvation induction.)

- Müller I, et al. *Succinate complex crystal structures of ... AtsK: steric aspects of enzyme self-hydroxylation*. J Mol Biol, 2005.

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(Self-hydroxylation protection.)